Unit 6, Lesson 18

Use an Algorithm with Partial Quotients

Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

Instructional Routines

- Number Talk

Goals

- Determine (orally and in writing) computation errors when dividing four-digit numbers by one-digit divisors and explain how to correct the errors.
- Use an algorithm with partial quotients to divide up to four-digit numbers by one-digit divisors.

Student Facing Learning Goals

Let's find quotients using an algorithm that uses partial quotients.

Lesson Purpose

The purpose of this lesson is for students to use an algorithm with partial quotients to find whole number quotients with up to four-digit dividends and one-digit divisors.

Narrative

In this lesson, students deepen and apply what they learned about partial quotients to divide four-digit numbers by single-digit divisors. They also analyze some common errors when using an algorithm with partial quotients. The last activity is optional and allows students to consider different ways to approach using partial quotients to divide.

Access for Students with Disabilities

- Representation

Access for English Learners

- MLR2

Lesson Timeline

Warm-up	10 min
Activity 1	15 min
Activity 2	10 min
Activity 3	10 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

Students have encountered many different ways to reason about division in the past few lessons. They might have other unique ways to think about division. Identify one or more ways in which your students' thinking offered a new insight or a positive surprise today.

Warm-up

10 min

Number Talk: Partial Quotients

Standards

Addressing **4.NBT.B.6**

Instructional Routines

- Number Talk

The purpose of this *Number Talk* is to elicit strategies and understandings students have for division using partial quotients. These understandings help students develop fluency and will be helpful later in this lesson when they interpret algorithm notation that shows partial quotients.

Student Task Statement

Find the value of each expression mentally.

- $500 \div 5$
- $60 \div 5$
- $5 \div 5$
- $565 \div 5$

Student Response

- 100: $5 \times 100 = 500$
- 12: $5 \times 12 = 60$
- 1: $5 \times 1 = 5$
- 113: Since $500 + 60 + 5 = 565$, I can add all the partial quotients from the first 3 problems.

Launch

- Display one expression.
- "Give me a signal when you have an answer and can explain how you got it."
- 1 minute: quiet think time

Activity

- Record answers and strategy.
- Keep expressions and work displayed.
- Repeat with each expression.

Activity Synthesis

Consider asking:

- "Who can restate _____'s reasoning in a different way?"
- "Did anyone have the same strategy but would explain it differently?"
- "Did anyone approach the problem in a different way?"
- "Does anyone want to add on to _____'s strategy?"

Activity 1

15 min

A Stack of Partial Quotients

Standards

Addressing **4.NBT.B.6, 4.OA.A.3**

The purpose of this activity is for students to develop an understanding of the vertical method of recording partial

quotients and use it to divide. In the *Launch*, students look at an algorithm that uses partial quotients and annotate it to show the multiplication that takes place. Then students find the value of several division expressions.

Access for English Language Learners

MLR2 Collect and Display. Synthesis: Direct attention to words collected and displayed from the previous lessons. Invite students to borrow language from the display as needed, and update it throughout the lesson.
Advances: Conversing, Reading

Student Task Statement

Jada uses partial quotients to find out how many groups of 7 are in 392.

1. Analyze Jada's steps in the partial-quotients algorithm.

```
  56
   9
   7
  40
7)392
− 280
 112
−  49
  63
−  63
   0
```

   a. Look at the 3 numbers above 392. What do they represent?

   b. Look at the 3 subtractions below 392. What do they represent?

2. Show another way you can decompose 392 to divide by 7.

3. Use a partial-quotients algorithm to find the value of $702 \div 3$.

Student Response

1. Sample response:
   a. First we make 7 groups of 40, then 7 groups of 7, and finally 9 groups of 7, for a total of 56 groups of 7.
   b. First, we subtract 7 groups of 40 (or 280) from the dividend, 392, leaving 112. Then we subtract 7 groups of 7 (or 49) from the 109, leaving 60. Finally we subtract 9 groups of 7 (or 63) leaving 4. The quotient is $40 + 7 + 9$, which is 56.

Launch

- Groups of 2
- Display Jada's work.
- "Talk with your partner about Jada's work and be ready to share your answers to the first question with the class."
- 2–3 minutes: partner discussion

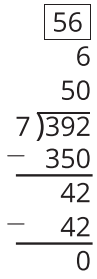
Activity

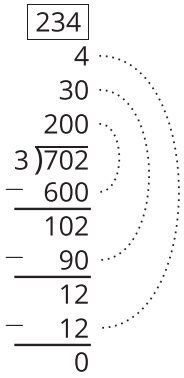
- Invite students to share their answers to parts a and b. As students explain what each of the values being subtracted represents, write a multiplication expression to represent each partial quotient. (7×40, 7×7, and 7×9)
- "Complete the rest of the problems on your own."
- 5–6 minutes: independent work time

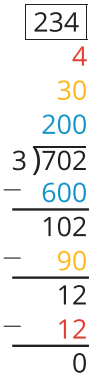
Activity Synthesis

- Display responses that show variation in how students decomposed 392.
- Highlight that there are countless ways of using partial quotients to divide a number, but it may be more efficient to divide larger groups than smaller groups. (For example, removing 50 groups of 7 or 350 once, instead of removing 70 five times. Removing smaller groups is just as valid.)
- Invite students to share their answers and reasoning for the last problem. To help students see the relationship between corresponding partial products and partial quotients, consider illustrating them visually. Examples:

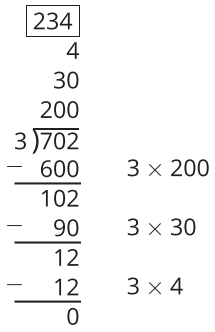
2. Sample response:

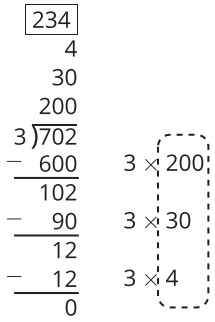






3. 234. Sample response:





Activity 2

10 min

Andre and Elena's Work

Standards

Addressing **4.NBT.B.6**

The purpose of this activity is for students to apply their understanding of partial quotients and the vertical recording method to divide four-digit numbers. They also identify some errors that are common when finding quotients this way. When students determine where the errors are and correct them, they critique the reasoning of others and construct viable arguments (MP3).

Access for Students with Disabilities

Representation: Access for Perception. Provide access to colored pencils. As shown in the previous activity, invite students to shade corresponding partial products and partial quotients.
Supports accessibility for: Visual-Spatial Processing, Organization

Student Task Statement

Andre and Elena divide 2,315 by 5. Before they begin, Andre says, "I can already tell that the quotient is going to be less than 500."

1. Decide if you agree with Andre without doing

Launch

- Groups of 2

Activity

- 1 minute: quiet think time for the first question

any calculations. Explain your reasoning.

2. Here is Andre and Elena's work. Each student made one or more errors. Identify the errors each student made. Then show a correct computation.

Andre's Work	Elena's Work
103 3 60 40 5) 2,315 −2,000 315 − 300 15 − 15 0	400 60 100 300 5) 2,315 −1,500 815 − 500 315 − 300 15

Student Response

Sample responses:

1. Yes, I agree with Andre. $500 \times 5 = 2500$ and that is more than the number we are dividing.
2.
   - Andre subtracted 2,000, which should've been 5×400, but he wrote 40 instead. The quotient should've been $400 + 60 + 3$, or 463.
   - Elena didn't finish the division. She stopped at 15, which can still be divided by 5 to get a partial quotient of 3.
   - Students show the quotient is 463 using partial quotients

- Briefly discuss student responses.
- 3 minutes: independent work time for the second question.
- 1–2 minutes: partner discussion

Activity Synthesis

- Select students to share their responses to the last question.
- "What are some ways to check our answers and avoid the mistakes Andre and Elena made? For example, how can we tell if dividing 2,315 by 5 gives a result in the 100s or in the 400s?" (We can estimate by multiplying the result by 5: 5×103 is a little over 500, and 5×400 is 2,000.)
- To reinforce the importance of keeping track of the partial quotients during division, consider annotating a corrected version of Elena's algorithm. Record the product that corresponds to each number being subtracted from the dividend (1,500, 500, 300, and 15).

403	
3	
60	
100	
300	
5) 2,315	
−1,500	5×300
815	
− 500	5×100
315	
− 300	5×60
15	
− 15	5×3
0	

Activity 3: Optional

10 min

Incomplete Calculations

Standards

Addressing **4.NBT.B.6**

The purpose of this activity is to reinforce the idea that there are many ways to use partial quotients to divide numbers. Students also see that some strategies are more practical or efficient than others.

Student Task Statement

Here are 4 calculations to find the value of $3{,}294 \div 3$, but each one is unfinished.

Complete at least 2 of the unfinished calculations.

A

```
      90
   1,000
3 )3,294
  −3,000    3 × 1,000
     294
   − 270    3 × 90
```

B

```
      80
     200
     400
     400
3 )3,294
  −1,200    3 × 400
   2,094
  −1,200    3 × 400
     894
   − 600    3 × 200
     294
   − 240    3 × 80
```

C

$600 \div 3 =$
$600 \div 3 =$
$600 \div 3 =$
$600 \div 3 =$
$600 \div 3 =$
$270 \div 3 =$

D

$3{,}300 \div 3 = 1{,}100$
$- \quad 6 \div 3 = \quad 2$

Student Response

A

```
 1,098
     8
    90
 1,000
3 )3,294
  −3,000    3 × 1,000
     294
   − 270    3 × 90
      24
   −  24    3 × 8
       0
```

B

```
 1,098
     8
    10
    80
   200
   400
   400
3 )3,294
  −1,200    3 × 400
   2,094
  −1,200    3 × 400
     894
   − 600    3 × 200
     294
   − 240    3 × 80
      54
   −  30    3 × 10
      24
   −  24    3 × 8
       0
```

Launch

- Groups of 2–4
- "Choose at least two calculations to finish. Make sure each calculation is completed by someone in your group."

Activity

- 3–4 minutes: independent work time
- 2 minutes: group discussion

Activity Synthesis

- Compare and contrast the four calculations. "How are the four strategies alike? How are they different?" (The first three are similar in that they involve partial quotients. The last one involves estimation.)
- "Which strategy or strategies do you find easy to follow? Hard to follow?"
- "Which strategy seems the most efficient? The least efficient?" (Calculations B and C seem lengthy and could be shortened by using larger multiples of 3. The last strategy seems most efficient.)

C

$$\begin{aligned} 600 \div 3 &= 200 \\ 600 \div 3 &= 200 \\ 600 \div 3 &= 200 \\ 600 \div 3 &= 200 \\ 600 \div 3 &= 200 \\ 270 \div 3 &= 90 \\ 24 \div 3 &= 8 \\ \hline 3{,}294 \div 3 &= 1{,}098 \end{aligned}$$

D

$$\begin{aligned} 3{,}300 \div 3 &= 1{,}100 \\ - \quad 6 \div 3 &= 2 \\ \hline 3{,}294 \div 3 &= 1{,}098 \end{aligned}$$

Lesson Synthesis

"Today we looked at different ways to divide multi-digit numbers by one-digit divisors. We saw estimating as a rather efficient way to find a quotient. How might we use estimation to find $5{,}970 \div 3$ or $6{,}986 \div 7$?" (For $5{,}970 \div 3$, notice that $5{,}970$ is 30 less than $6{,}000$, which is $3 \times 2{,}000$. Thirty is 3×10, so $5{,}970$ is $3 \times 1{,}990$. For $6{,}986 \div 7$, notice that $6{,}986$ is close to $7{,}000$, which is $7 \times 1{,}000$, and $6{,}986$ is 14 or 7×2 less than $7{,}000$. So $6{,}986$ is 7×998.)

"How can we check the result of our division to make sure it's not off?" (We can multiply the result by the divisor, adding the remainder if there is one, and see if it gives the dividend.)

Suggested Centers

- Compare (1–5), Stage 4: Divide within 100 with One-Digit Divisors (Supporting)
- Rolling for Fractions (3–5), Stage 2: Multiply a Fraction by a Whole Number (Supporting)

Cool-down

5 min

Find a Quotient

Standards

Addressing **4.NBT.B.6**

Student Task Statement

Find the value of $2{,}023 \div 7$. Use partial quotients to show your reasoning.

Student Response

289. Sample response: partial quotients 200, 80, 9

Responding to Student Thinking

Students make computation errors resulting from addition, subtraction, or multiplication fluency.

Next Day Supports
Before the *Warm-up*, invite students to work in small groups to discuss a correct response to this *Cool-down*.